

A Novel Model of the Spread of Emerald Ash Borers in Pennsylvania

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What is Emerald Ash Borer (EAB)?



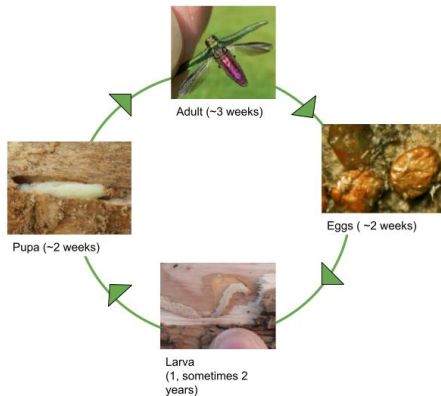
Adult EAB (Source: [11])

EAB is **invasive**.

The native range of EAB is in East Asia, where it inhabits **already-diseased and dying** ash trees.

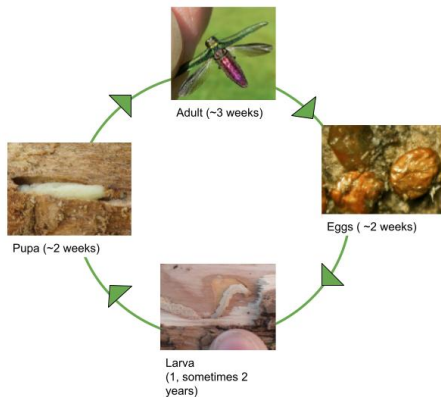
In the US, EAB infests **healthy** trees. In some cases, nearly all the ash trees in a region are **dead within 6 years** of EAB introduction.

Lifecycle of EAB



Images sourced from [12]

Lifecycle of EAB



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Larval “galleries” (Source: [14])

EAB lays eggs on ash tree bark. The larva hatch and tunnel into the tree, where they eat the inner bark.

This is what kills the trees.

Modes of Spread

Natural Spread: Adult EAB typically **fly** up to half a mile from their hatch site.



Infested Firewood (Source: [20])

Modes of Spread

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Human-Assisted Spread: EAB can be introduced to uninfested regions by the **transportation of infested lumber and firewood**



Infested Firewood (Source: [20])

Why do Ash Trees Matter?

Ash trees are...



Source:[5]

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- a **canopy species**, providing shade for native plant species



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Why do Ash Trees Matter?

Ash trees are...

- a **canopy species**, providing shade for native plant species
- a source of **food** for many animals
- a major source of **lumber** in the US
- important to many Indigenous nations



Source:[5]

Goals of Modeling EAB

Model the spread of EAB across a region, incorporating stochastics to model the randomness of individual beetle movement

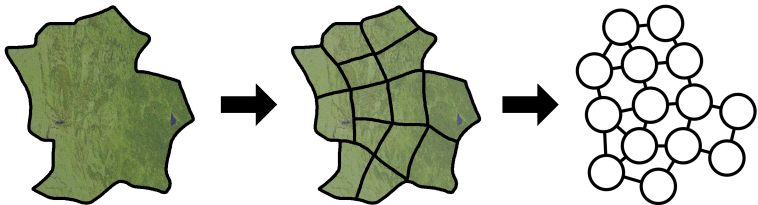
Goals of Modeling EAB

Model the spread of EAB across a region, incorporating stochastics to model the randomness of individual beetle movement

Investigate the impact of human actions on the spread of EAB and resulting ash tree death

Model Overview

We model a region by representing it with a graph, where each node represents a small area, and each edge represents the adjacency of these subdivided areas.



Local Dynamics

We describe the dynamics within each node with the following dynamical system (X = ash trees, Y = EAB):

Dynamical System (Full)

$$X' = \rho X \left(1 - \frac{X}{K} \right) - \alpha X \ln(1 + Y)$$

$$Y' = \beta Y \left(1 - \frac{Y}{QX} \right) - O(Y) + I(Y)$$

where $\rho = 0.1, \alpha = 0.05, \beta = 3, Q = 9000, K = \text{variable}$

Local Dynamics

Dynamical System (Illustrative)

$$X' = A(X) - B(X, Y)$$

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- $B(X, Y)$ describes the detrimental effects of the beetles on the trees
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- $O(Y)$ and $I(Y)$ describe the number of beetles moving out of and in to the node

Discretization of Local Dynamics

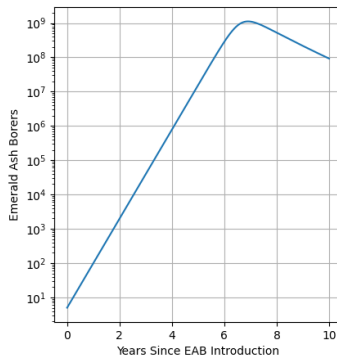
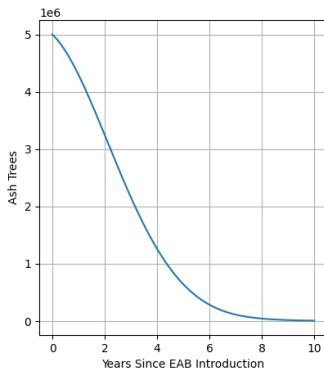
The $O(Y)$ and $I(Y)$ are non-deterministic, so we use the forward Euler method to discretize our model (with time step Δt , allowing us to deal with the stochastic terms:

Dynamical System (Discretized)

$$X_{t+1} = X_t + \Delta t \left[\rho X_t \left(1 - \frac{X_t}{K} \right) - \alpha X_t \ln(1 + Y_t) \right]$$

$$Y_{t+1} = Y_t + \Delta t \left[\beta Y_t \left(1 - \frac{Y_t}{QX_t} \right) - O(Y_t) + I(Y_t) \right]$$

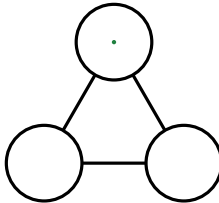
Graph of Local Dynamics



Graph of system with $K = X_0 = 5000000$ and $Y_0 = 5$ for a duration of 10 years (with $O = I = 0$)

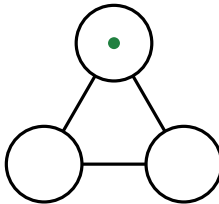
Natural Transregional Spread

For simplicity, we model each node as a perfect circle, beetles start at the center of a node and spread outwards.



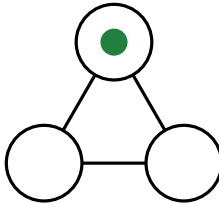
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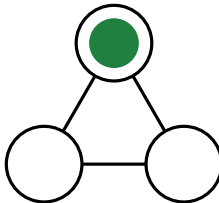
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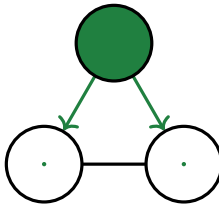
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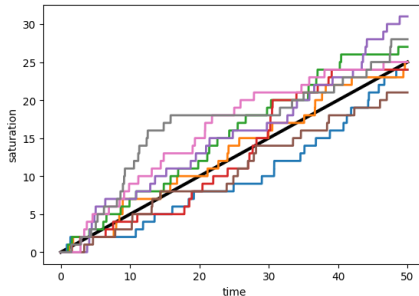
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Beetle Travel Within A Node

For each node, we keep track of its radius and its “beetle saturation”.
Then, on each simulation step:

$$\Delta \text{saturation}(t) = \begin{cases} +1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$



Beetle Travel Between Nodes

When saturation = radius, we “turn on” the beetle exodus term $O(Y)$:

Number of beetles leaving a node at time t

$$O(Y_t) = \begin{cases} 0 & \text{when saturation} < \text{radius} \\ \eta Y_t & \text{when saturation} = \text{radius} \end{cases}$$

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The beetles which leave a node N goes to its neighbours:

Number of beetles entering a node at time t (part 1)

$$I_1(Y_{Nt}) = \sum_{M \in \text{adjacencies of } N} \frac{O(Y_{Mt})}{\text{number of adjacencies of } M}$$

Human-Driven Transregional Spread

Last but not least, humans bring EAB between nodes in infested lumber.

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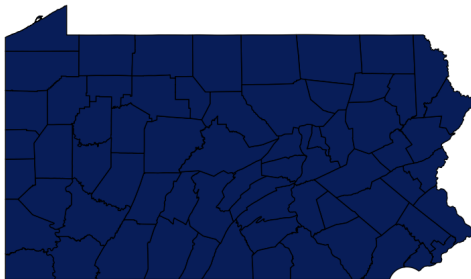
Let $N(t)$ be the number of infested nodes at time t , then:

Number of beetles entering a node at time t (part 2)

$$I_2(Y_t) = \begin{cases} 10 & \text{with probability } qN(t)\Delta t \\ 0 & \text{with probability } 1 - qN(t)\Delta t \end{cases}$$

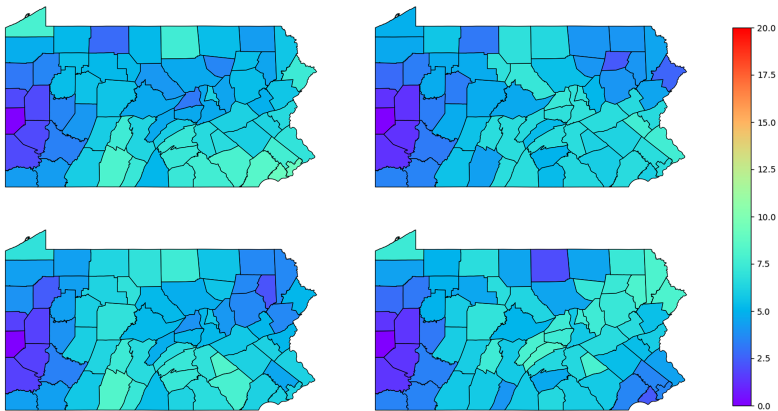
Animation of Model

t = year 0.0



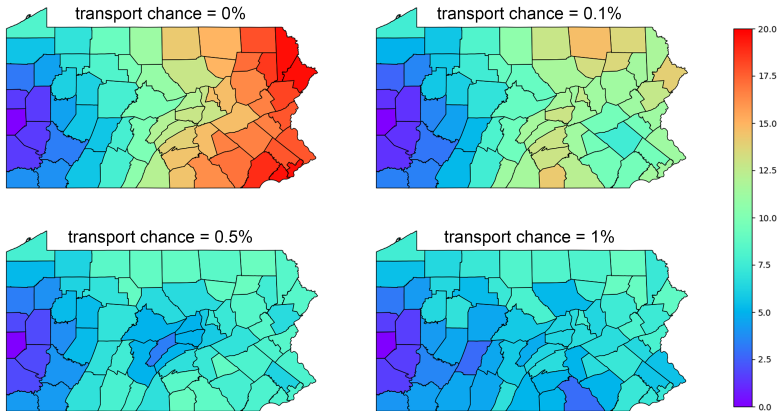
Animation of the model, where dark blue represents trees at 100% carrying capacity and light yellow represents a lack of trees.

Results



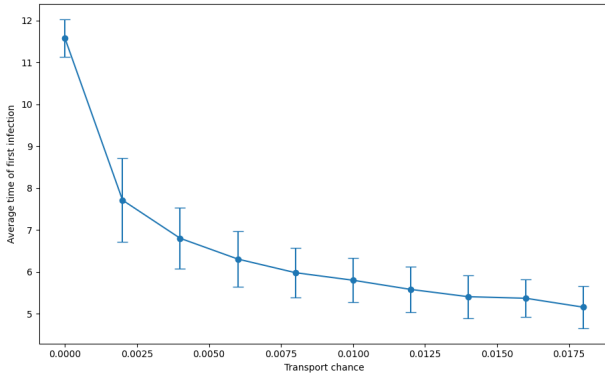
Several simulations with 1% chance of human-driven transport, color-coded by the number of years after initial infection it takes for EAB to appear.

Results



Several simulations with difference chances of human-driven transport, color-coded by the number of years after initial infection it takes for EAB to appear.

Results



The mean and standard deviation of the mean time of first infection for select transport probabilities between 0% to 2% (200 trials for Monte Carlo method)

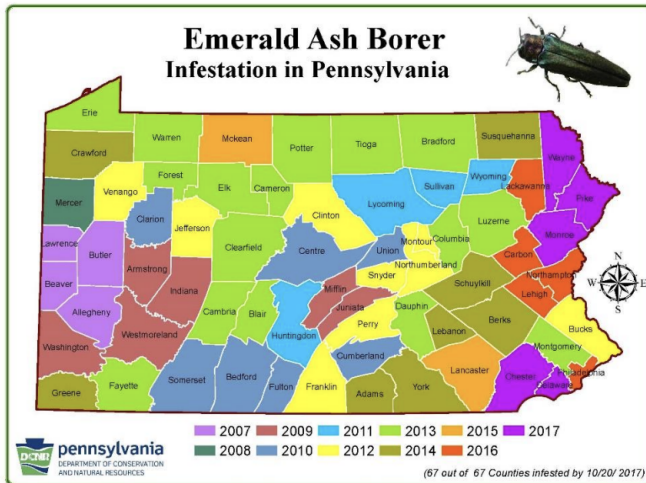
Discussion

With no human transport, mean time to first infection around 11.5 years.

With human transport, mean time to first infection around 9 (0.1% chance), 6.5 (0.5% chance), or 6 (1% chance) years.

The 1% chance of transport best matches real-world data.

Discussion



Map of Pennsylvania counties with year EAB was first detected. (Source: [18])

Model Limitations

Geographic Assumptions

- Counties are circles
- Constant tree density in every county

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Parameter Assumptions

- Same human transport chance to and from every node
- Not great data for α , the death rate of trees once infected, or ρ , the tree growth rate
- Doesn't account for other interventions (pesticides, parasitoids)

Future Work

Future researchers could expand on this model by:

- Decreasing node size
- Modeling accurate tree density of each forest
- Adjusting transportation probabilities based on frequency of human travel and campground locations
- Adding terms for other interventions

Conclusion

Eventually, EAB will infect every node regardless of human-driven transportation

Human actions can increase or delay the spread

What can we do? Keep firewood local and spread the word

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